For Examiner's Use

5 An ideal transformer is connected to a sinusoidal a.c. supply, as shown in Fig. 5.1. The primary coil has a r.m.s. current of 0.85 A.

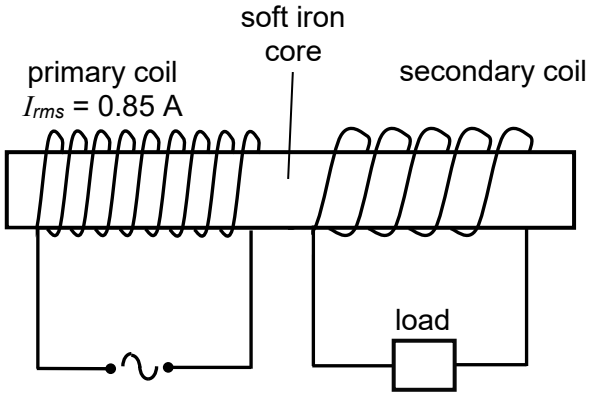

Fig. 5.1

(a) Use the laws of electromagnetic induction to explain how a potential difference can be developed across the secondary coil.

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……..….. [3]

(b) **(i)** The primary coil contains 9 turns per cm, calculate the maximum magnetic field strength at its centre.

maximum magnetic field strength = …… T [2]

(ii) The ratio of the number of turns in the primary to secondary coil is 16 : 1.

Calculate the peak current in the load.

peak current = ... A [2]

(iii) The variation with time *t* of current I_p in the primary coil is shown in Fig. 5.2.

On Fig. 5.3, sketch the variation with time *t* of current I_s in the secondary coil. [1]

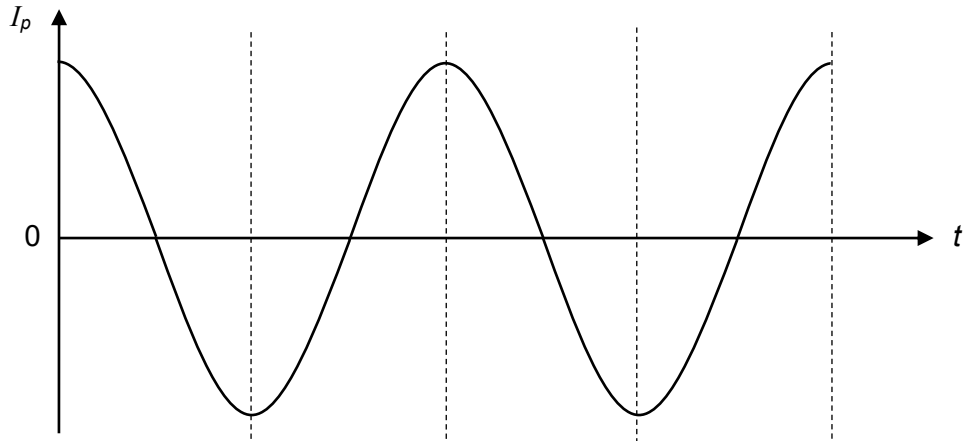

Fig. 5.2

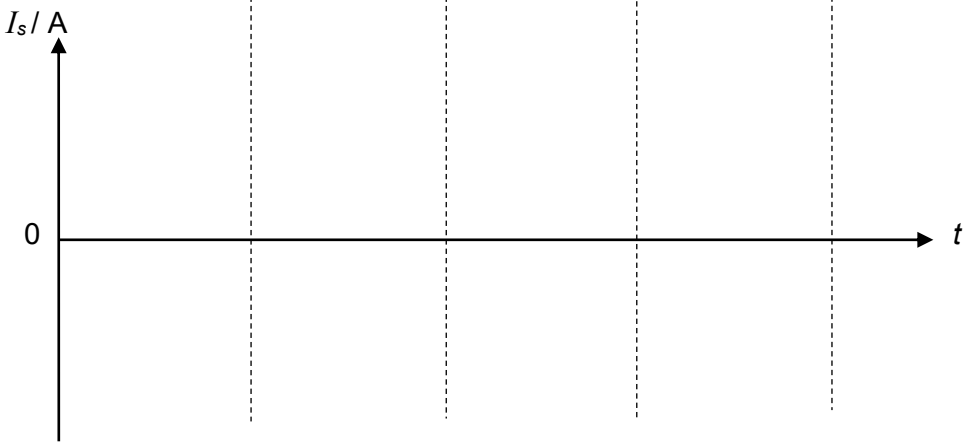

Fig. 5.3

(iv) State and explain how the answer to **b (iii)** will change if the iron core is removed from the transformer.

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